

Systems Biology 2025/26

Spatiotemporal Signal Propagation

Deadline: May 12, 2026, 18:00
Submission: GitHub repository + PDF report
Total points: 10 points

Introduction

This project builds directly on materials you worked with during tutorials:

- **Exploratory notebooks** (`spatiotemporal-starter/students-notebooks/`): data loading, temporal dynamics, spatial structure, collective behavior
- **Analysis scripts** (`spatiotemporal-starter/scripts/` and `spatiotemporal-continuation/scripts/`): `spatiotemporal_signal_propagation.py` (single-block analysis) and `compare_spatiotemporal_behavior.py` (multi-block comparison)
- **Mathematical handout & worksheet** (`spatiotemporal-continuation/notes/`): graph formalization, exposure metrics, Relative Risk (RR), propagation analysis framework

Additional prepared materials for this project (located in the repository root):

- **notebooks-part2/**: Example workflows for running scripts, parameter sensitivity analysis, mutation comparisons. You will use these to generate analysis outputs.
- **notebooks-part3/**: Methodological extensions-lagged exposure ($RR^{(\tau)}$), global thresholds, dose-response curves

Your task is to apply and extend these methods to answer research questions about cell-to-cell signal propagation. The project has two parts:

- **Part A (5 points)**: Guided analysis modifying existing scripts with clear deliverables
- **Part B (5 points)**: Independent research: choose structured questions or formulate your own

All analyses must be reproducible. Modify existing workflows rather than writing from scratch. Emphasis is on *biological interpretation* and *methodological decisions*, not just code execution.

Part A: Guided Analysis (5 points)

Task A1: Multi-Mutation Spatiotemporal Comparison (2 points)

Research Question: Do different PI3K-AKT pathway mutations alter the strength of spatiotemporal signal propagation?

Methods: Modify and run `compare_spatiotemporal_behavior.py` (from `spatiotemporal-continuation/scripts/`) to analyze all five cell lines: (WT, AKT1_E17K, PIK3CA_E545K, PIK3CA_H1047R, PTEN_del) using default parameters (`spatial_radius=60`, `future_window_frames=3`, `signal_col='ERKKTR_ratio'`). Extract `mean_relative_risk` from `group_level_summary.csv`. Perform pairwise statistical comparisons (Mann–Whitney U test with Bonferroni correction) between WT and each mutant.

Deliverables & Grading (2 pts):

- [0.5 pt] Table: `mutations_comparison_table.csv` (mutation, mean_RR, std_error, p_value_vs_WT, significant)
- [0.5 pt] Statistical tests correctly applied with multiple comparison correction
- [0.5 pt] Bar plot showing mean RR $\pm$ SE with significance markers (* $p < 0.05$)
- [0.5 pt] Interpretation (150–250 words): Which mutations differ? Propose biological mechanism.

Reference: See `notebooks-part2/Part1_Block3_Comparison.ipynb` for example workflow.

Task A2: Lagged Exposure Analysis Across Mutations (2 points)

Research Question: Do different mutations exhibit different spatiotemporal relay timescales (how quickly does activation propagate from neighbor to focal cell)?

Methods: Adapt `notebooks-part3/Part2_L1_LaggedExposure.ipynb` to compute $RR^{(\tau)}$ for lags $\tau = 0, 1, 2, 3, 4, 5, 6$ frames (0, 5, 10, 15, 20, 25, 30 min). Perform this for WT and *two* mutant cell lines of your choice (justify selection: e.g., "AKT1_E17K and PTEN_del, both hyperactivate AKT via different mechanisms"). For each cell line, identify optimal lag τ^* where $RR^{(\tau)}$ is maximized.

Deliverables & Grading (2 pts):

- [0.5 pt] Correct $RR^{(\tau)}$ computation for three cell lines across all lags
- [0.5 pt] Line plot: $RR^{(\tau)}$ vs. τ for WT + two mutants, with markers at τ^*
- [0.5 pt] Summary table: mutation, optimal_lag_tau_star, max_RR
- [0.5 pt] Interpretation (150–250 words): Do propagation timescales differ? If $\tau^* > 0$ for one mutation but $\tau^* = 0$ for another, what does this imply about signaling dynamics?

Reference: `notebooks-part3/Part2_L1_LaggedExposure.ipynb` demonstrates lagged exposure framework.

Task A3: Parameter Robustness Assessment (1 point)

Research Question: How sensitive is the Relative Risk metric to analysis parameter choices?

Methods: First, generate parameter sweeps by running cells 2, 5, 11, and 14 in `notebooks-part2/Part1_Block2_ParameterSensitivity.ipynb` (takes ≈ 10 minutes total; each cell runs the analysis script with different parameter values). This creates sweep folders in `analysis_outputs/`. Then extract `mean_relative_risk` from `summary.json` files in each sweep subfolder (choose either spatial radius sweep: `sweep_r*/` OR future window sweep: `sweep_w*/`). Plot RR vs. parameter value. Recommend optimal parameter based on stability and biological interpretability.

Deliverables & Grading (1 pt):

- [0.25 pt] Successfully generated parameter sweeps (verify output folders exist)
- [0.25 pt] Line/scatter plot: RR vs. parameter value with data points
- [0.5 pt] Recommendation (100–150 words): Which parameter value? Justify via statistical robustness and biological reasoning.

Reference: Follow workflow in `notebooks-part2/Part1_Block2_ParameterSensitivity.ipynb`.

Part B: Independent Research (5 points)

Choose ONE of the following options:

Option 1: Structured Research Question (maximum 3 points)

Select *one* of the three questions below. Each has methodological hints but requires you to implement the analysis, interpret results, and draw biological conclusions.

Question B1: ERK vs. AKT Propagation Comparison

Hypothesis: ERK and AKT biosensors report on different signaling pathways that may exhibit distinct spatiotemporal propagation patterns.

Methods Hint: Run `spatiotemporal_signal_propagation.py` for both `signal_col='ERKKTR_ratio'` and `signal_col='FoxO3A_ratio'` on the same experiment-site blocks. Compare RR values, jump thresholds (θ), and optionally lagged exposure curves $RR^{(\tau)}$.

Deliverables & Grading (1–3 pts):

- [1 pt] Basic: RR values for both biosensors, simple comparative plot
- [2 pts] + Statistical test (Mann–Whitney U) comparing RR distributions + interpretation of biological differences
- [3 pts] + Advanced analysis (lag comparison, mutation-specific patterns) + strong biological reasoning linking to pathway cross-talk

Question B2: Spatial Heterogeneity Analysis

Hypothesis: Cells at the edge of the field-of-view may exhibit different propagation behavior than central cells due to boundary effects or lower neighbor counts.

Methods Hint: Use `objNuclei_Location_Center_X/Y` to classify cells as "edge" vs. "central" (e.g., compute distance from field-of-view centroid; classify outer 20% as edge). Stratify nodes table by category and compute RR_{edge} and RR_{central} separately.

Deliverables & Grading (1–3 pts):

- [1 pt] Basic: Correct spatial classification and cell counts per category

- [2 pts] + Comparison table (spatial_category, n_cells, mean_neighbors, RR) + statistical test (Mann–Whitney U)
- [3 pts] + Thoughtful interpretation: Why might edge effects matter? Implications for in vitro vs. in vivo signaling.

Question B3: Dose-Response Extension

Hypothesis: The intensity of neighbor signal (not just binary "jumping" vs. "not jumping") may better predict focal cell's future jump, revealing dose-dependent relay mechanism.

Methods Hint: Adapt `notebooks-part3/Part2_L3_DoseResponse.ipynb`. Use `neighbor_mean_signal` as continuous exposure variable. Bin cells into exposure quartiles (or quintiles) based on neighbor signal intensity. Compute RR for each bin:

$$RR_{\text{bin}} = \frac{p_{\text{jump}|\text{bin}}}{p_{\text{jump}|\text{baseline}}}$$

where baseline is lowest exposure bin. Test for trend.

Deliverables & Grading (1–3 pts):

- [1 pt] Basic: Correct binning and RR computation per bin
- [2 pts] + Dose-response plot (RR vs. mean neighbor signal) + trend test (Spearman correlation or linear regression)
- [3 pts] + Biological interpretation of dose-response shape (threshold effect? saturation?) + mechanistic hypothesis

Option 2: Open Research Question (maximum 5 points)

Formulate and solve an original research question using the spatiotemporal propagation framework. Your question must be clearly stated, require distinct analysis approach (not simple tutorial replication), use graph-based propagation framework (nodes, edges, exposure, RR) or extend it meaningfully, and include quantitative assessment.

Examples of acceptable scope: *Multi-step propagation* (do second-degree neighbors contribute beyond first-degree?); *Oscillation frequency* (do WT vs. mutants show different inter-jump intervals? correlation with local RR?); *Local density modulation* (does RR depend on absolute neighbor count?); *Directional propagation* (detect directional bias using spatial edge orientations).

Required Components & Grading (5 pts):

- [1 pt] Introduction: Clear hypothesis, biological motivation, and research question (1 paragraph)
- [2 pts] Methods & Analysis: Describe data processing, metric definitions, workflow (1–2 paragraphs). Analysis method implemented correctly (code works, logic sound).
- [1 pt] Results: Findings with ≥ 1 high-quality plot, quantitative summary (table or statistical test)
- [1 pt] Discussion: Biological interpretation + limitations or alternative explanations (1–2 paragraphs)

Submission Guidelines

GitHub Repository Structure

Your repository should contain (or at least be analogous to) the following structure, with clear organization and documentation to ensure reproducibility:

```
repo-root/
  README.md                # Setup + reproduction instructions
  notebooks/
    TaskA1_MutationComparison.ipynb
    TaskA2_LaggedExposure.ipynb
    TaskA3_ParameterRobustness.ipynb
    TaskB_IndependentResearch.ipynb
  outputs/
    mutations_comparison_table.csv
    mutations_barplot.png
    lagged_exposure_table.csv
    [additional outputs for Part B]
  scripts/                  # Modified scripts (if applicable)
```

README.md requirements:

- Brief project description (2–3 sentences)
- Environment setup: Python version, package installation from `requirements.txt`
- Step-by-step reproduction commands for each task
- List of generated output files

PDF Report

Maximum length: 4–5 pages (excluding references and title page)

Required structure:

1. **Title Page:** Project title, your name, student ID, submission date
2. **Part A: Guided Analysis** (2–3 pages)

For each task (A1, A2, A3):

- Brief methods summary (1 paragraph, reference specific notebooks)
- Key results: include main table or figure
- Interpretation (1–2 paragraphs addressing the research question)

3. **Part B: Independent Research** (1–2 pages)

Structure as a mini research report:

- *Introduction:* Research question, hypothesis, biological motivation
- *Methods:* Data processing, analysis approach
- *Results:* Main findings with visualization
- *Discussion:* Biological interpretation, limitations, future directions

Code Quality Expectations

Your code will be evaluated on three criteria:

- **Clarity:** Use functions (not just sequential cells), meaningful variable names, and comments explaining non-obvious analytical steps
- **Reproducibility:** All analyses must run successfully from README instructions without errors or missing dependencies
- **Organization:** Logical notebook structure following the pattern:
`imports` → `load data` → `analysis` → `visualization` → `interpretation`

Grading Summary

Component	Points	Key Deliverables
Code Quality & Reproducibility	-2-0	README, organized notebooks, comments
- Unclear structure & functions	-1	
- No instruction and reproducibility trace in README	-1	
Part A: Guided Analysis	5	
Task A1: Mutation Comparison	2	Table, plot, stats, interpretation
Task A2: Lagged Exposure	2	RR(τ) curves, τ^* table, interpretation
Task A3: Parameter Robustness	1	Robustness plot, recommendation
Part B: Independent Research	3-5	
Option 1: Structured Question	max 3	Analysis, visualization, interpretation
Option 2: Open Research	max 5	Hypothesis, methods, results, discussion
TOTAL	10	

Good luck, and I look forward to seeing your creative approaches to spatiotemporal signal propagation analysis!